

Canonical notation, identifiers, and glossary

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This file is the single source of truth for mathematical notation, code identifiers, and controlled vocabulary across the manuscripts under `analysis/report/`. New manuscripts should adopt it rather than re-deriving symbols, and any deviation should be stated explicitly in the deviating paper's Methods section.

The audit behind it is `analysis/report/whitepaper-notation-audit.md`, whose PDF binds this file in as its appendix so that the two cannot drift apart.

Symbols and identifiers listed here were extracted from the manuscript sources, from R/, from the drivers under `analysis/scripts/`, and from the archived `.rds` summaries.

Part 1. Mathematical symbols

Indices and structure

Symbol	Meaning
i	participant index
t	timepoint index; weekly measurement occasions throughout
j, ij	period or occasion index within participant (papers 04, 06)
N	total number of participants in the trial, across all randomization paths; never a per-path count. See Part 2, sample size and randomization paths
P	number of randomization paths in a design; the expected allocation to each is N/P
$n_{\text{reps}}, n_{\text{sim}}$	Monte Carlo replicates per cell
k	cycles per participant in design sweeps; also the Weibull shape parameter in the decay family

Outcome and latent components

Symbol	Meaning
Y_{it}	observed symptom score; the modelled outcome (code column <code>Sx</code>)
BL_i	participant-specific baseline level
BR_{it}	biological (pharmacological) response component
PB_{it}	placebo-belief response component
TV_{it}	time-variant natural-history component
ε_{it}	observation-level residual
u_i	participant random intercept
η	expectancy covariate (measured belief), paper 06

Treatment exposure and carryover

Symbol	Meaning
D_{it}	binary drug state: 1 on drug, 0 off drug
$D_{bc,it}$	continuous exposure-decayed drug indicator: 1 on drug, $e^{-\lambda t_{sd}}$ off drug (code column <code>Dbc</code>)

Symbol	Meaning
t_{sd}	time since discontinuation; input to the carryover decay (code <code>tsd</code>)
t_{od}	cumulative time on drug; input to the Gompertz response (code <code>tod</code>)
t_{pb}	time-by-placebo-belief accumulation (code <code>tpb</code>)
$t_{1/2}$	carryover half-life (code <code>carryover_t1half</code>)
λ	carryover decay rate, $\ln 2/t_{1/2}$ (code <code>lambda_cor</code>)
λ_w	Weibull rate, $(\ln 2)^{1/k}/t_{1/2}$

Decay families implemented on the data-generating side: exponential $e^{-\lambda t}$; linear $\max(0, 1 - t/(2t_{1/2}))$; Weibull $e^{-(\lambda_w t)^k}$; power $\max(0, (1 - t/(3t_{1/2}))^p)$.

Biomarker and moderation

Symbol	Meaning
B_i	participant-level pre-treatment biomarker value (code <code>bm</code>)
b_i	standardized biomarker, $(B_i - \bar{B})/s_B$ (code <code>bm_z</code>)
c_{bm}	moderation parameter of the covariance-moderation architecture: on-drug correlation between biomarker and BR . A correlation, bounded in $[-1, 1]$
β_{bm}	moderation parameter of the mean-moderation architecture: dimensionless multiplier scaling the treatment effect by b_i
$c_{bm,A}, c_{bm,B}$	independent mean-channel and covariance-channel weights of the dual-channel architecture
σ_{BR}, σ_{bm}	standard deviations of the response component and of the biomarker

Estimand, inference, and performance

Symbol	Meaning
$\beta_{bm:D}$	the target estimand: the biomarker-by-treatment interaction coefficient (<code>bm:Dbc</code>)
$\beta_{bm}^{BR}, \beta_{bm}^{PB}, \beta_{bm}^{TV}$	component-specific biomarker slopes (paper 06)
$\beta_{bm}^{\text{lumped}}$	the one-component (single-indicator) biomarker slope
β_D	treatment main effect (papers 04, 05)
$\theta, \hat{\theta}$	a generic interaction coefficient; used only where the argument is deliberately not N-of-1 specific (paper 10)
θ_{true}	calibrated true value, $-c_{bm} \sigma_{BR}$
ρ	AR(1) within-factor serial correlation
κ	ratio of mean model-based standard error to empirical standard deviation; $\kappa > 1$ means a conservative test
δ	standardized bias of the test statistic
ν	denominator degrees of freedom
α	nominal significance level; 0.05 throughout
π	power, $\Pr(p < \alpha)$
γ	latent-class gating slope (paper 03); a $BR \times PB$ interaction coefficient (paper 06)

Symbol	Meaning
Z_i	unobserved latent class label (paper 03)

The five rules the symbols encode

1. D_{it} **is binary**; $D_{bc,it}$ **is continuous**. These are different objects, and the distinction is the subject of the carryover-specification manuscript. Never use one for the other.
2. c_{bm} **is a correlation**; β_{bm} **is a multiplier**. They are calibrated to a common numeric scale, because the mean-channel shift is applied as $\beta_{bm} b_i \sigma_{BR}$ on the standardized biomarker, so the reference value 0.45 denotes a matched moderation strength under either architecture. The symbols are nonetheless not interchangeable. A paper reporting both architectures at a matched value may use one shared label, but must say so.
3. **Mathematics in model definitions, code identifiers in code contexts**. Write $Y_{ij} = \beta_0 + \beta_D D_{bc,ij} + \dots$ for the model; write `Sx ~ bm + t + Dbc + bm:Dbc` in `\texttt{}` when quoting the R formula. Never mix the two inside one expression.
4. **N is the total sample size**. N is the total number of participants across all randomization paths, and the expected allocation to each of P paths is N/P : with $P = 4$ and $N = 140$, each path receives about 35. Never use N for a per-path count. A paper may name a per-path count alongside the total, in words or as N/P , but must give N itself as the total. Because two of the drivers fix the per-path count instead, the reported N cannot be taken on trust: check the driver against the table in Part 2 before quoting a sample size.
5. **State the sign convention**. The three components are non-negative *reductions* in symptom severity, so an increase in any component lowers Y . Treatment effects and interaction coefficients are consequently negative; moderation parameters are positive.

What the linter checks

`tools/notation-lint.pl` mechanizes the part of this file that can be checked from the manuscript sources. Run it over the manuscripts:

```
perl tools/notation-lint.pl analysis/report/*/report.Rmd
```

It skips fenced R chunks and `verbatim` blocks and strips `\texttt{}` and backtick spans, so it sees reader-facing prose and mathematics only. Four checks are implemented.

Check	Fires on	Rule
<code>bare-Dbc</code>	<code>Dbc</code> as a symbol outside a code span	1
<code>bare-Sx</code>	<code>Sx</code> as a symbol outside a code span	3
<code>legacy-spec-label</code>	<code>A1</code> , <code>A2</code> , <code>A3</code>	Part 3, specifications
<code>N-per-path</code>	a value of N directly qualified as per-path	4

`N-per-path` fires on `$N = 35$ per randomization path` but not on `$N = 140$ (70 per path)`, which is the compliant way to give both numbers. A day-denominated $t_{1/2}$ is exempted when the file carries a `**Units.**` paragraph, which is how rule 3's units requirement is enforced.

Rule 2 and rule 5 are not mechanized: whether a shared c_{bm} label is declared, and whether the sign convention is stated, both need a reader. One known false positive stands, a `bare-Sx` inside a backtick span that crosses a line break in paper 02.

Units

Carryover half-life $t_{1/2}$ is quoted in **weeks** in the interaction-focused manuscripts, on the canonical grid $\{0, 0.5, 1.0\}$, and in **days** in the main-effect and test-procedure manuscripts, on a pharmacokinetic scale with a 3-day baseline. Both conventions are retained because each paper's figures are drawn on its own scale. Every paper using days carries an explicit units paragraph giving the conversion (1 week = 7 days). Timepoints t are weekly in both conventions.

Part 2. Code identifiers and stored data values

These are the names as they appear in R/, in the drivers, and in the archived .rds summaries.

Where a stored value is also a reporting label, the two are kept **identical**, so that no display-time mapping stands between the data and the page. A mapping layer is a standing hazard: it has to be applied at every display site, it fails silently when a rename reaches one layer and not the other, and it makes a stored table unreadable without the manuscript beside it. Where a stored value is a package API argument (`dgp_architecture = 'mvn'`), it stays as the API defines it and the manuscript maps it to a reporting name, which is the ordinary factor-level-to-label relationship.

Exported package functions

`buildtrialdesign`, `buildSigma`, `generateData`, `censordata`, `lme_analysis`, `generateSimulatedResults`, `validateParameterGrid`, `cumulative`, `modgompertz`, `trajectoryShape`, `shape_logistic`, `shape_hyperbolic_tangent`, `shape_piecewise_linear_breakpoint`, `characterize_carryover`, `analyze_trial_extended`, `print_carryover_summary`, `print_trial_summary`, `plotfactortrajectories`, `PlotModelingResults`, `reknitsimresults`.

Principal arguments

Identifier	Role
<code>modelparam</code>	list: sample size <code>N</code> , moderation <code>c.bm</code> . The meaning of <code>N</code> depends on the caller; see sample size and randomization paths below
<code>respparam</code> <code>blparam</code>	Gompertz parameters <code>maxr</code> , <code>rate</code> , <code>disp</code> per component biomarker and nuisance means and standard deviations
<code>trialdesign</code>	output of <code>buildtrialdesign</code> : <code>timepoints</code> , <code>timeptnames</code> , <code>expectancies</code> , <code>ondrug</code>
<code>dgp_architecture</code> <code>carryover_t1half</code>	'mvn', 'mean_moderation', or 'combined' carryover half-life
<code>lambda_cor</code> <code>br_family</code> <code>br_p2</code> , <code>br_p3</code>	decay rate; derived from the half-life when NA response-curve family: 'gompertz' and alternatives shape parameters of the alternative families
<code>moderation_scaling</code> <code>cached_sigma</code>	'constant' or trajectory-scaled pre-built covariance matrix, reused across parallel workers
<code>makePositiveDefinite</code> <code>n_cores</code> <code>save_chunks</code>	positive-definiteness repair switch worker count; -1 auto-detects progressive checkpointing for long runs

Sample size and randomization paths

Rule 4 fixes N as the total. The complication is that `modelparam$N` does not mean the same thing at every level of the call stack, and two families of driver read it differently.

The two levels. `generateData` generates one path: it takes a single `trialdesign` and treats N as the count for that path alone. `generateSimulatedResults` generates a whole design: it takes N as the total and allocates it, `Ns <- N %/% nP` with the remainder spread one per path. A driver that loops over `td$trialpaths` itself is therefore working at the lower level and must divide the total before the loop. A driver that passes the same full N to every iteration generates $N \times P$ participants, and its reported N is a per-path count wearing the wrong name.

Path counts. P is a property of the design, not a free parameter.

Design	P	Paths
OL	1	on drug throughout
CO	2	drug-then-placebo, placebo-then-drug
OL+BDC	2	discontinuation at either of two visits
Hybrid	4	two discontinuation points crossed with two crossover orders

Paper 06 also uses a reduced single-path OL+BDC variant for its smaller cells and a two-path sixteen-visit variant for its larger ones, so P changes within that paper’s sample-size axis.

Which drivers allocate and which duplicate. Established by inspection of every driver reachable from a current manuscript, on 2026-08-05.

Driver	Reads N as	Mechanism
R/generateSimulatedResults.R	total	<code>Ns <- N %/% nP</code> plus remainder
analysis/scripts/carryover-sensitivity/simulation-core.R	total	<code>generate_data_multi_path, floor(N / n_paths)</code>
implementations/noflpower/R/simulation.R	total	same integer split
analysis/scripts/test-procedure-design-sensitivity/01-study1-test-procedure.R	total	explicit <code>split_N</code>
analysis/scripts/quick-sim/*.R	per path	hand-rolled for <code>(g in seq_along(paths))</code> passing one mp
analysis/scripts/component-decomposition/*.R	per path	same shape
analysis/scripts/gompertz-evaluation/02-faithful-trajectory-sweep.R	per path	same shape

Status by paper. The manuscripts report totals throughout, and as of 2026-08-08 the drivers and stored output for papers 01, 06, 07 and 09 were brought into line with them. One entry is weaker than the others: the driver behind paper 01’s Section 3.3 robustness run was not located, so its per-path reading is taken from that paper’s own statement of its convention rather than from inspected code.

Paper	Driver was	Now
01	per path	driver allocates the total; re-run at a matched $N = 70$ for all three designs
02, 05, 08, 10, 11 03, 04	total coincident (single-path or path-free DGP)	already correct; untouched unaffected
06	per path	all drivers allocate the total; stored N migrated to totals; see the caveat below
07	per path	driver allocates the total; re-run; stored N added as 70
09	per path	driver allocates the total; re-run at a matched $N = 70$ for all four designs

How the corrections were validated. The work ran in two stages, and it is worth keeping them apart. The first stage corrected the label only. Re-expressing a grid in totals and allocating across paths leaves the per-path counts unchanged, so the generated data should be identical and nothing but the recorded N should move. That was checked before each re-run by generating one replicate both ways and comparing: for every driver converted, the drawn data and the fitted coefficients were identical. The re-runs then reproduced the published cell power values exactly, in all 36 cells of paper 01, all 16 of paper 07 and all 16 of paper 09, with per-replicate coefficients moving by at most 7×10^{-9} (paper 01), 0 (paper 07) and 4×10^{-5} (paper 09), which is the convergence-tolerance noise on identical data.

The second stage, for papers 01 and 09 only, changed the sample sizes themselves to match the designs, so it was expected to change results and did. Its internal check is that the cells whose totals were already 70 must still reproduce exactly while the others move: in paper 01 the CO and OL+BDC cells came back bit-identical and only the Hybrid column changed, by up to 0.233 in power.

Paper 06 was migrated, not re-run. Study A alone holds 200,000 stored fits, so re-running it to change a column that the equivalence check proves is mislabeled would cost hours to reproduce the same numbers. Instead `analysis/scripts/component-decomposition/migrate-n-to-total.R` recoded the stored N in place, 109 files across five output directories, with pre-migration copies in `analysis/.n-migration-backup/` and read-back verification of row counts and of the absence of legacy values. A column-by-column comparison against the backups confirms that nothing but N changed. The mapping is per directory because the studies differ in path count: the study-A and contamination grids switch to the two-path design above a threshold, so 100 and 150 became 200 and 300 while 35 and 70 were already single-path totals; study-B recovery is two-path throughout, so 70 and 150 became 140 and 300; study-B balanced is single-path and was left alone.

All four paper-06 drivers are now converted: `run-study-a-prod.R`, `run-study-contam.R` and `run-study-b-recovery.R` read N as a total and allocate it, and `run-study-b-balanced.R` needed nothing because its balanced-placebo design has a single path. Each conversion was checked two ways. An equivalence gate regenerated one replicate per design type under both parameterizations and got identical coefficients, standard errors, p-values and formula-dropped status. Then every stored checkpoint was matched against the new grid's N for its `cell_id`: 18 of 18 for the contamination study and 8 of 8 for study-B recovery, no mismatches, so each driver agrees with the data already migrated on disk. Where a fit function takes an N argument it is passed the per-path count, which is what it carried before, so the formula ladders keyed on it are unaffected. Study-B recovery's three fits declare the argument but never read it; it is kept per-path there for consistency with the other drivers.

The substantive problem was fixed on 2026-08-08. Fixing the per-path count across designs with different P leaves a comparison that is not matched on N , so papers 01 and 09 were re-run with every design at a common total of $N = 70$. This changes numbers rather than labels, and it changed conclusions in both papers.

In paper 01 only the Hybrid column moved, since CO and OL+BDC were already at 70 and reproduced exactly. Hybrid power fell from about 0.99 to 0.84, which took it off the ceiling and roughly tripled its measured carryover loss: 1.4 to 6.9 percent under mean moderation and 5.1 to 15.4 percent under covariance moderation. The paper's headline architecture contrast survives, with the difference-of-differences z at Hybrid moving from 3.96 to 2.80 and OL+BDC unchanged at 7.64, but its claim that mean-moderation losses are uniformly 1 to 3 percent did not, and the Hybrid design is no longer saturated anywhere in the grid.

In paper 09 the OL and hybrid rows moved. The happy-accident ordering held, hybrid 0.968 above OL+BDC 0.950 above crossover 0.854 above OL 0.088 at every dropout pattern, but the hybrid margin over OL+BDC collapsed to within one MCSE, and the paper's second conclusion, that the most powerful design is also the most dropout-robust, was withdrawn: hybrid loss under 40 percent biased dropout rose from 0.4 to 5.8 percentage points, level with OL+BDC's 5.6, so the earlier near-immunity was a ceiling effect. The OL row also stopped coinciding with α , moving from 0.022-0.046 to 0.086-0.112, which showed that reading it as a null-calibration check had been a numerical coincidence at one sample size rather than a property of the design.

The lesson worth carrying: an unmatched design comparison does not merely inflate one design, it can manufacture a robustness finding out of a ceiling effect.

Data columns

ptID (participant), Sx (outcome, the Y_{it} above), bm (biomarker), bm_z (standardized biomarker), Dbc (continuous exposure indicator, $D_{bc,it}$), tsd, tod, tpb, t_wk (time in weeks), path (randomization path).

Stored factor levels and their reporting labels

Column	Stored values	Reported as	Mapped?
spec	E1, E2, E3	E1, E2, E3	no, identical
carryover_form	exponential, linear, weibull	as stored	no, identical
design	CO, Hybrid, OLBDC	CO, Hybrid, OL+BDC	only OLBDC to OL+BDC
dgp_arch	mean_moderation, mvn, combined	Mean, Covariance, Dual	yes; these are package API values

The spec column stored A1, A2, A3 until 2026-07-31. Twenty-one .rds files were migrated to E1/E2/E3 and the producing drivers updated to emit them, so the display-time mapping that previously bridged the two has been retired. Pre-migration copies are in analysis/.spec-migration-backup/. Output archived before that date, including anything under a paper's own archive/, still uses the old values.

Summary-table columns

n_reps, n_converged, converged_frac, non_convergence_rate, power, bias, mean_estimate, true_value, empirical_se, mse, coverage, and an mc_se_-prefixed Monte Carlo standard error for each (mc_se_power, mc_se_bias, mc_se_mse, mc_se_coverage, mc_se_non_convergence).

Part 3. Controlled vocabulary and glossary

The project lexicon at docs/29-nof1-precision-medicine-lexicon.md holds the full terminology set (241 terms across 13 thematic sections, with cross-field synonyms). The entries below are the subset carrying a fixed, enforced meaning in this compendium.

Trial designs

OL (open-label). Unblinded titration; no off-drug contrast, so it supplies no within-subject interaction information. **BDC** (blinded discontinuation). A blinded switch to placebo after open-label response. **OL+BDC**. Open-label titration followed by blinded discontinuation. **CO** (crossover). Traditional two-period within-subject crossover. **Hybrid**. The Hendrickson (2020) design: open-label titration, blinded discontinuation, then a brief crossover. The programme's reference design. Capitalize it. **Aggregated N-of-1 trial**. Many single-patient on/off series pooled into one mixed model.

Define each abbreviation at first use in every paper. Each design's randomization-path count is given in Part 2, under sample size and randomization paths, and is not repeated here.

Data-generating architectures

Named by the channel that carries the interaction. Single letters are no longer used.

Name in prose	Legends and tables	Data value	Former label
the mean-moderation architecture	Mean	mean_moderation	Architecture A
the covariance-moderation architecture	Covariance	mvn	Architecture B

Name in prose	Legends and tables	Data value	Former label
the dual-channel architecture	Dual	combined	Architecture C

Mean-moderation. The biomarker scales the treatment effect in the conditional mean; the interaction lives in the first moment. The near-universal convention in the trial-simulation literature. **Covariance-moderation.** The biomarker-response correlation is treatment-state dependent; the interaction lives in the second moment. The Hendrickson formulation, and the one whose power is most eroded by carryover. **Dual-channel.** Both channels active with independent weights $c_{bm,A}$ and $c_{bm,B}$; the other two are its single-channel limits. A data-generating construct only: the two weights are not separately identifiable from a fitted `bm:Dbc` coefficient.

Analysis specifications for carryover

E1. Binary on-drug indicator; carryover ignored. **E2.** Exposure-weighted continuous predictor D_{bc} , committing to an assumed decay form and half-life. **E3.** Binary indicator plus a lagged just-off-drug nuisance term; the classical Jones-Kenward crossover device, assuming no decay form.

The letter E stands for the exposure regressor that distinguishes them. These were labeled A1/A2/A3 until 2026-07-31; the stored data and the drivers were migrated to E1/E2/E3, so the labels now agree everywhere and nothing is mapped at display time.

Why not A/B/C and M1/M2/M3. The original scheme used A/B/C for architectures and A1/A2/A3 for specifications, colliding on the letter A within a single manuscript. Relabeling the specifications M1/M2/M3 removed that collision but created a second one: the calibration manuscript already uses M0-M3 for its working-covariance model ladder, and those labels are keys into archived data. E1/E2/E3 is free across the whole corpus, and naming the architectures by mechanism removes the class of collision entirely.

Response-curve families

Modified Gompertz (default), $y = \maxr \cdot \exp(-\text{disp} \cdot \exp(-\text{rate} \cdot t))$ with a vertical offset so the curve passes through the origin. Alternatives: **logistic** (symmetric sigmoidal), **hyperbolic tangent** (fast-saturating), **piecewise-linear breakpoint** (no smoothness assumption).

Missing data and dropout

MCAR, MAR, MNAR in the standard Rubin sense. **Informative dropout.** Hazard depending on cumulative symptom worsening since baseline. **Happy accident.** The randomization-path selection effect by which informative dropout preferentially preserves the most informative crossover blocks.

Inference

LME. Linear mixed-effects model; the programme’s default analysis. **corCAR1.** Continuous-time AR(1) residual correlation structure. **GEE.** Generalized estimating equations; the marginal comparator. **Mancl-DeRouen, CR2.** Small-sample bias-corrected sandwich variance estimators. **RM-ANOVA.** Repeated-measures analysis of variance; the strict form requires a dichotomized biomarker. **Anti-conservative.** Realized Type I error above nominal. **Working covariance.** The assumed residual covariance of the LME; reserve **working correlation structure** for the GEE object.

Estimands and reporting

Biomarker-treatment interaction. The programme’s core estimand; equivalently the patient-by-treatment interaction, or heterogeneity of treatment effects (HTE) in the precision-medicine literature. **Predictive biomarker.** Predicts differential benefit, as opposed to a **prognostic** biomarker, which predicts outcome regardless of treatment. **ADEMP.** The Morris, White and Crowther (2019) reporting framework: aims, data-generating mechanisms, estimands, methods, performance measures. **MCSE.** Monte Carlo standard error; reported with every performance measure.

Enumeration conventions

Simulation studies. Number them (Study 1, Study 2, ...) rather than lettering them. **Sensitivity sweeps.** Labels S1, S2, ... are paper-local and are not comparable across papers. Cite them with the paper prefix when crossing manuscripts (05-S2, not S2).

Prose terminology

Symbol conventions are settled here; prose terminology is settled by `docs/30-terminology-consistency-plan.md`, executed 2026-07-30 with its D1 spelling decision locked to **US English**, overriding that document's original frequency-based recommendation of British forms. The two are complementary: this file governs mathematics and identifiers, that one governs surface forms and acronym expansion.

Rendered on 2026-08-08 at 21:18 PDT. Source: `~/prj/res/36-pmsimstats-ng/pmsimstats-ng/analysis/report/NOTATION.md`